pyhydra: an open, modular Python library for reproducible hydroclimatic modelling and probabilistic flood-risk analysis

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Abstract

Flood-risk studies chain data acquisition, extreme-value and copula statistics, stochastic rainfall generation, climate bias correction and hydrological and hydraulic simulation, usually through ad-hoc scripts that are hard to reuse and reproduce. **pyhydra** is an open, modular Python library that consolidates this chain into fifteen sub-modules: harmonised access to public hydro-climatic data, statistical and copula tools, stochastic generators, bias correction and downscaling, automation of HEC-HMS, HEC-RAS, SFINCS and SWAT+, and uncertainty propagation, including a pilot-based rule that decides whether an expensive solver can be emulated within a fixed simulation budget. It is illustrated through four workflows at three Spanish sites; external-engine outputs are archived and third-party adoption is still to be shown. **pyhydra** is GPL-3.0-licensed, installable from GitHub (not yet on PyPI) and archived on Zenodo; the companion platform HYDRA exposes its notebooks in a browser.

Keywords: flood risk, reproducible research, hydroclimatic data harmonisation, stochastic rainfall generation, extreme value statistics, uncertainty quantification

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Required Metadata

Current code version

Nr.	Code metadata description	Metadata
C1	Current code version	v0.3.1
C2	Permanent GitHub link to code/repository used for this code version	`https://github.com/navass11/pyhydra/tree/v0.3.1`; archived at `https://doi.org/10.5281/zenodo.23599934`
C3	Legal Code License	GNU General Public License v3.0 only (GPL-3.0-only)
C4	Code versioning system used	git
C5	Software code languages, tools, and services used	Python ≥3.9; continuously tested on Python 3.9–3.12 (section 2.2)
C6	Compilation requirements, operating environments & dependencies	Not yet published on PyPI; install with `pip` directly from the GitHub repository in C2, optionally pinned to the tag in C1 (section 2.1). Core dependencies are declared in `pyproject.toml`; optional functionality is distributed through the `statistics`, `geospatial`, `models`, `notebooks` and `all` extras (section 2.1). External engines (HEC-HMS, HEC-RAS, SFINCS, SWAT+) are not bundled and must be installed on the host
C7	If available Link to developer documentation/manual	`https://github.com/navass11/pyhydra/tree/v0.3.1` (README, `docs/` package/installation guide, and `notebooks/pilot_cases/`, which include the four illustrative examples of section 3, inspectable and executable from this repository without HYDRA, subject to the external-engine and project-data requirements listed in Table 3). The companion platform HYDRA additionally exposes the same notebooks through a browser at `https://github.com/navass11/HYDRA/tree/v0.2.2`
C8	Support email for questions	salvador.navas@hidralab.com

Table 1: Code metadata (mandatory)

1. Motivation and significance

Flood-risk assessment combines heterogeneous hydro-climatic data, extreme-value and dependence models, scenario generation, climate bias correction,

and hydrological/hydraulic simulation. Study-specific scripts make this chain difficult to reproduce or transfer, particularly where resources for characterising flood hazard are limited [1].

pyhydra consolidates a line of impact-based flood-risk research into reusable, versioned Python components. The founding Besaya study combined geostatistical event generation and probabilistic regression to estimate return periods directly on hydraulic impacts [2]. Subsequent applications covered spatial rainfall generation with NEOPRENE [3], Gaussian-copula/MaxDiss flood-frequency analysis in Madrid [4], climate impacts on reservoir inflows [5] and Andean hydropower [6], and vine-copula design storms [7]. These studies repeatedly implemented overlapping statistical and modelling components.

The contribution is an architecture that composes established methods from source data to impact-based frequency products. Its four blocks provide data access, climate/statistical analysis, model integration and uncertainty quantification (UQ). Statistical components work independently of hydraulic models, while adapters retain external engines as replaceable components: HEC-HMS [8], HEC-RAS [9], SFINCS [10] and SWAT+ [11]. Public data services remain independent sources [12–15]. The companion platform HYDRA adds browser access and a Docker environment; it is separate from the library described in table 1.

Related frameworks address complementary tasks: CLIMADA combines hazards, exposure and impact functions [16]; RainyDay generates storms by transposing remotely sensed fields [17]; and HydroMT/wflow supports reproducible distributed-model construction and execution [18, 19]. **pyhydra** combines multivariate extremes, stochastic generation, climate correction, model adapters and impact-frequency analysis within one workflow, using ecosystem packages such as HydroMT-SFINCS where appropriate.

2. Software description

2.1. Software architecture

pyhydra is a pip-installable Python package organised into four blocks: data sources, climate and statistics, model integration, and UQ (fig. 1). Exchange formats include `pandas.DataFrame`, `xarray.Dataset`, GeoTIFF and `geopandas.GeoDataFrame`, allowing components to be replaced or bypassed. The library contains no web infrastructure. HYDRA adds JupyterLab, FastAPI, an Astro/nginx frontend, Docker and Azure Container Apps deployment. **pyhydra** incorporates GPL-v3 code and is distributed under GPL-3.0-only; HYDRA is independently licensed under MIT.

Sub-modules expose their public API through `__init__.py`, using `from pyhydra.<block>.<module> import <name>`. OpenTURNS, a core dependency, supplies Gaussian-copula fitting. Dependencies are imported lazily, so components remain usable without unrelated heavy packages. Four extras group optional dependencies: `statistics` (PyMC, pyvinecopulib, lmoments3), `geospatial` (GeoPandas, rasterio, pykrige, earthaccess, hydromt_sfincs), `models` (hecdss, pySWATPlus, spotpy, NEOPRENE) and `notebooks` (seaborn, contextily, xlrd, openpyxl). A legacy `geo` extra pro-

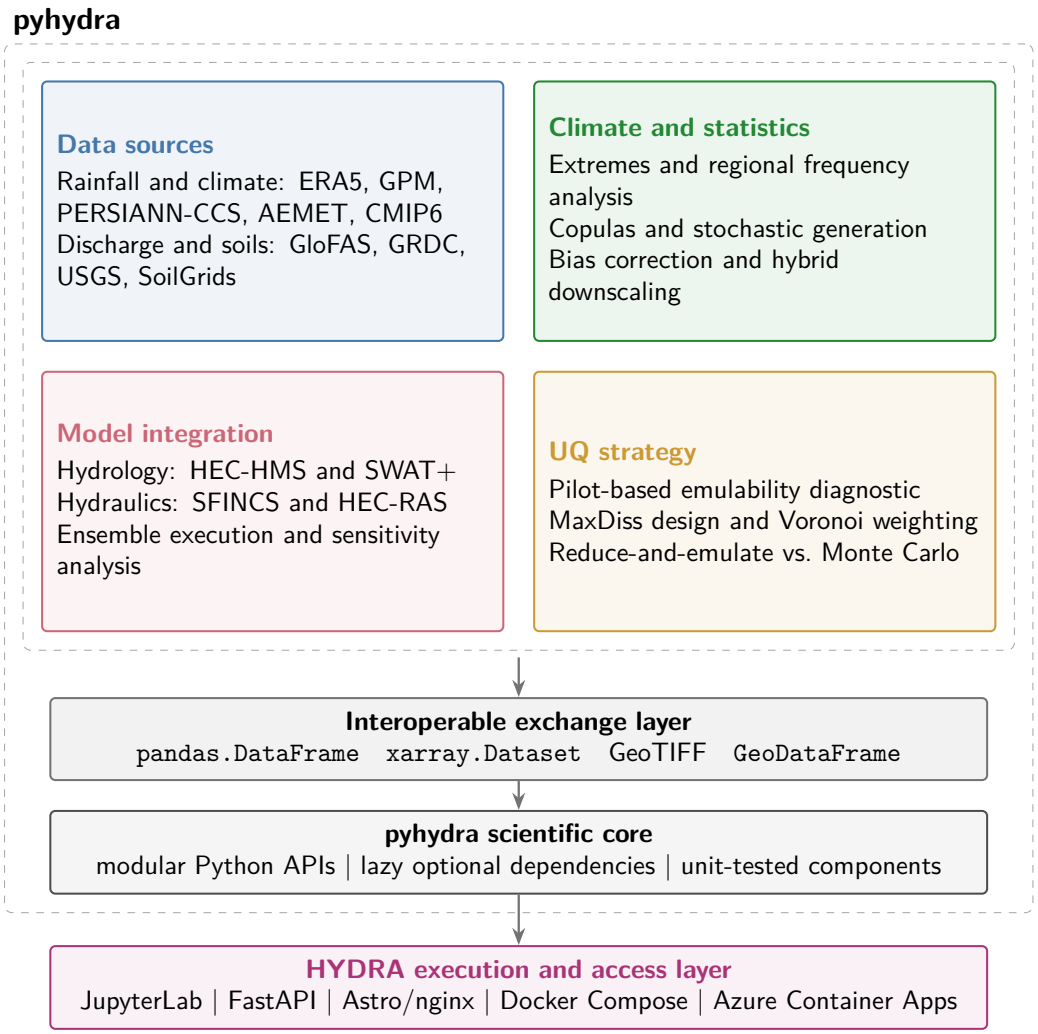


Figure 1: Software architecture. **pyhydra** provides four functional blocks of reusable Python components connected through standard exchange formats; HYDRA wraps the scientific core with reproducible execution, service and access layers. Colours identify functional roles and do not encode values.

vides GDAL bindings; `all` covers all five. CoSMoS_py requires separate installation from its repository. HYDRA's Docker Compose stack supplies extended scientific dependencies and mounts project data from a persistent file share; GitHub Actions deploys the stack to Azure Container Apps.

2.2. Software functionalities

Table 2 summarises the fifteen sub-modules. They implement established bias-correction [20], index-flood/L-moments [21] and copula [22] methods. Modularity, explicit parameters and data conventions, standard exchange formats, documentation and notebooks support independent use, reproducibility, interoperability and transfer.

A `pytest` suite comprises 299 tests in 19 files. GitHub Actions runs it on Ubuntu across Python 3.9–3.12, with core and lightweight extended dependencies including PyMC. For commit `e96fbca` (23 October 2026), 299 tests pass on Python 3.10–3.12; on 3.9, 298 pass and one PyMC/pytensor test is skipped because the resolved stack is incompatible with NumPy. That test is guarded by `optional_dependency` when the stack is unavailable or incompatible. Line coverage is 32% (31% on 3.9), concentrated in statistical, modelling-adapter and UQ utilities. Data connectors and engine automation have lower coverage and are exercised, rather than formally verified, by pilot notebooks. See the archived CI run.

Reproducibility boundary. Statistical and bias-correction examples use included or synthetic data. Public-data downloads require service credentials and recorded queries; model workflows also require host-installed engines and project files. Some third-party inputs cannot be redistributed. The audit contract (fig. 2) records sources, parameters, seeds, engine configuration and environment; containerisation alone does not make restricted data or external executables available.

Uncertainty propagation and analysis. UQ includes asymptotic, bootstrap and Bayesian extreme-value estimates; regional posterior propagation and spatial prediction variance; and roughness ensembles aggregated into impact statistics with outlier screening. The `uq` layer chooses whether to emulate an expensive solver within budget `n_max`. A nested maximum-dissimilarity pilot, charged to that budget, measures cross-validated emulability against a declared emulator family. The layer then completes a reduce-and-emulate design or falls back to direct Monte Carlo. `StrategyDecision.margin` identifies borderline choices, while sequential design classifies added points as improving, converged or deteriorating.

Table 2: **pyhydra** modules by functional block.

Module	Block	Main capability
`data_sources.rainfall`	Data	ERA5, GPM, PERSIANN-CCS, AEMET, OGIMET, Meteostat
`data_sources.river_discharge`	Data	GloFAS, GRDC, USGS
`data_sources.climate_change`	Data	CMIP6 via CDS and ESGF
`data_sources.soils`	Data	SoilGrids download & USDA texture classification
`climate.time_series`	Climate	Event extraction, GEV/GPD fitting, NSE/KGE/PBIAS
`climate.spatial_analysis`	Climate	Regional frequency analysis, Gaussian copulas (OpenTURNS) / vine copulas (pyvinecopulib), IDW/kriging/GP interpolation
`climate.stochastic_generation`	Climate	NEOPRENE (point NSRP, spatial STNSRP), CoSMoS random fields
`climate.bias_correction`	Climate	Delta method, empirical/quantile-delta/scaled-distribution mapping
`climate.hybrid_downscaling`	Climate	Hydrograph classification, MaxDiss selection, k-NN/RBF flood-map reconstruction
`modeling.hydrology.hec_hms`	Modelling	Project generation, DSS I/O, SCE-UA calibration via spotpy
`modeling.hydrology.swat`	Modelling	SWAT+ input generation, execution, output parsing
`modeling.hydraulic.sfincs`	Modelling	Model setup, boundary conditions, batch execution
`modeling.hydraulic.hec_ras`	Modelling	1D/2D project automation, DSS result extraction
`modeling.hydraulic.sensitivity`	Modelling	Manning Monte Carlo ensembles, spatial impact regression
`uq`	UQ	Pilot-based emulability diagnostic, MaxDiss design, reduce-and-emulate vs. Monte Carlo

Abbreviations: GEV/GPD, Generalised Extreme Value / Generalised Pareto Distribution; NSE/KGE/PBIAS, Nash–Sutcliffe Efficiency / Kling–Gupta Efficiency / Percent Bias; IDW, Inverse Distance Weighting; GP, Gaussian Process; DSS, Data Storage System (HEC file format); SCE-UA, Shuffled Complex Evolution – University of Arizona; RBF, Radial Basis Function.

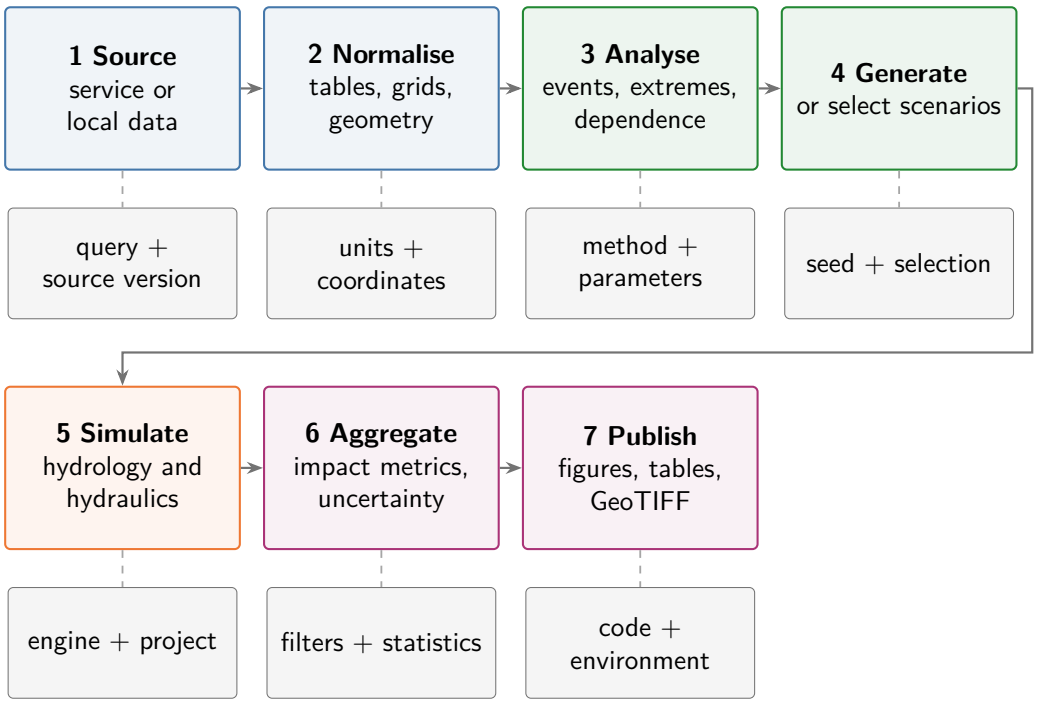


Figure 2: Reproducible workflow contract. Each transformation produces a scientific artefact and an audit record sufficient to identify its source, parameters, stochastic seed, external-engine configuration and software environment. Grey boxes: the audit record kept for each stage. Colours identify functional stages and do not encode values.

2.3. Sample code snippets

The snippet demonstrates three independently usable components:

```
from pyhydra.climate.bias_correction import BiasCorrection
from pyhydra.climate.stochastic_generation.point import NSRPModel
from pyhydra.climate.spatial_analysis.copulas import BivariateCopula

# 1. Bias-correct a regional climate model precipitation series
corrected = BiasCorrection.fit_transform(
    obs, hist, future, var='pr', method='qdm')

# 2. Calibrate and simulate a Neyman-Scott rectangular-pulse
# stochastic rainfall generator (NEOPRENE; 'models' extra)
model = NSRPModel(seasonality='monthly')
model.fit(observed_series)
synthetic = model.simulate(year_ini=2000, year_fin=2100)

# 3. OR-scenario 100-year iso-return-period contour of a compound
# rainfall-discharge event
copula = BivariateCopula(family='gumbel').fit(rain_max, discharge_max)
x_or, y_or = copula.joint_return_period(t=100, kind='or')
```

3. Illustrative examples

Four workflows at three Spanish sites are archived as 25 notebooks under `notebooks/pilot_cases/` in the versioned repository (table 1, C7), also accessible through HYDRA. Table 3 identifies inputs and execution requirements. All 25 finish on archived data: nine Besaya notebooks skip HEC-HMS/HEC-RAS steps when the engine is absent; the other 16 require no engine.

Los Corrales de Buelna (Besaya River, Cantabria). The founding case [2, 25] combines 42 years of continuous hydrological simulation (1970–2012), 13 rain gauges, geostatistical rainfall reconstruction, copula-generated events and k-NN flood-map reconstruction ($k = 6$, selected by cross-validation). Return periods are estimated on hydraulic impacts. Simulated floods may affect over 13% of the population and 20% of the town's area [25].

M-30 Manzanares (Madrid). The published study [4] classified 1,761 events from 17 gauges using PCA and k-means. A Gaussian copula generated approximately one million events; MaxDiss selected 625 for HEC-HMS and steady HEC-RAS 1D (303 cross-sections). k-NN depth reconstruction and a compound Poisson frequency model gave a 500-year depth of 7.61 m at

Puente de Toledo, 0.83 m above the univariate-GEV estimate (6.78 m). Because historical events and engine runs cannot be redistributed, the notebooks fit `FloodEventCopula` to an archived 10,000-event synthetic sample.

Table 3: Traceability of the four illustrative workflows. Notebook paths are relative to `https://github.com/navass11/pyhydra/tree/v0.3.1/notebooks/pilot_cases/`. "Yes" marks inputs that are the authors' own or openly licensed. "Partial" marks cases where some inputs are third-party records (AEMET, SIAR, AVAMET, CHC), whose reuse and redistribution follow their providers' terms of use, or are project-specific and not redistributed here.

Case	Path	Open data	External engine	Reproduction
Besaya	`los_corrales_buelna/` (9 nb.*)	Partial (projects and results on Zenodo [23]; gauge-derived series under provider terms)	HEC-HMS, HEC-RAS	Conditioned on the engines and on the gauge records
M-30 Manzanares	`m30_manzanares/` (6 nb.)	Partial (synthetic-catalogue sample and archived results on Zenodo [23])	None (engine results archived)	Copula generation and MaxDiss re-run with **pyhydra**; design depths read from archived engine results
Valencia DANA	`valencia_dana/` (3 nb.†)	Partial (AEMET, SIAR and AVAMET series under provider terms; Zenodo [23])	None	Reproducible for the point and regional (RFA) analyses with those records
SFINCS–HEC-RAS	`manning_rugosidades/` (7 nb.)	Yes (Zenodo dataset [24])	SFINCS, HEC-RAS	Ensemble aggregation re-runs from the archived 2×1,000 outputs; re-simulation needs the engines

*Steps 01–08 form the impact-frequency workflow; step 09 applies the `uq` decision layer to its MaxDiss design. †Steps 01–02 produce the results reported here; step 03 extends the regional analysis with a hierarchical Bayesian GEV (PyMC), using the legacy station matrix and the Stan reference results archived with the data.

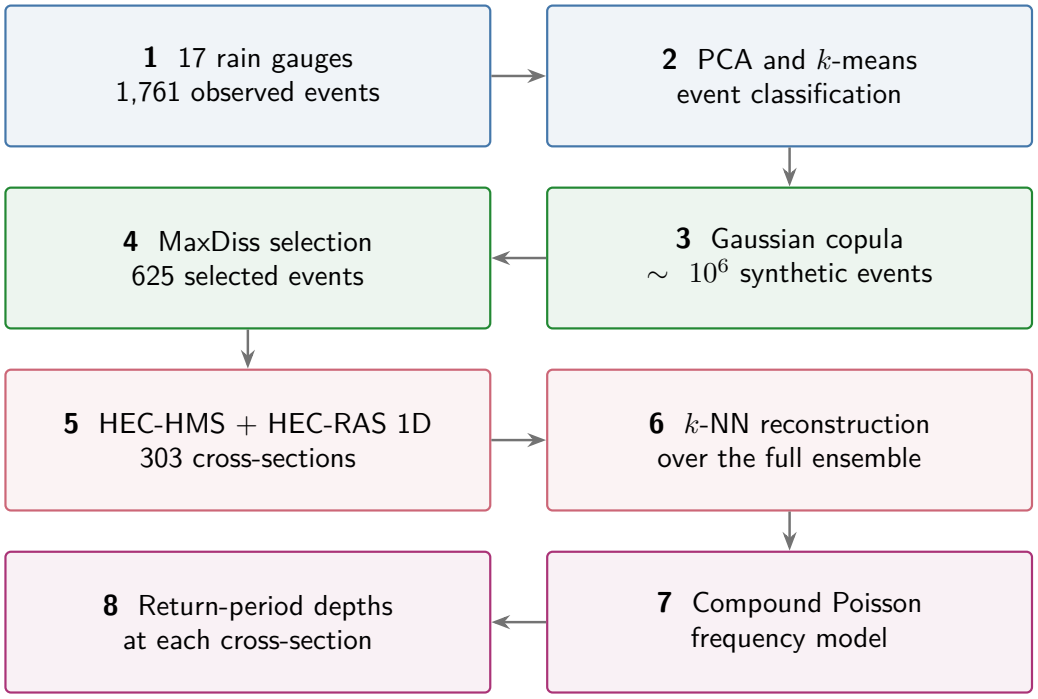


Figure 3: M-30 pipeline [4]. Notebooks re-run copula generation and MaxDiss; simulation, emulation and frequency outputs are archived. Blue: observations/classification; green: generation/selection; orange: simulation/emulation; purple: frequency outputs at Puente de Toledo and Represa nº9.

An OpenTURNS Normal copula with zero-inflated marginals regenerates 100,000 events over 84 variables, reproducing dry-gauge shares and rank dependence (mean $|\Delta\rho| = 0.011$ across 3,486 pairs). MaxDiss selects 625 events in that space. Published design depths are read from archived outputs, not recomputed (fig. 3).

Valencia's DANA (October 2024). Turís (8337X) recorded 710.8 mm on the calendar day of 29 October 2024, during the record-breaking event documented by AEMET [26]. Two notebooks, obtainable with `pyhydra-get-data valencia_dana` without run-time AEMET credentials, fit GEV distributions with and without the event. Point fits use MLE, L-moments and Bayesian/MCMC; regional frequency analysis (RFA) uses nine-station index-flood standardisation with MLE and L-moments. The estimated 100-year level rises from 218–271 to 756–1,241 mm, a factor of 3.5–4.6 (fig. 4 and table 4). All values are recomputed with v0.3.1. These short annual-maximum records have an upper tail dominated by one record. The ratios measure sensitivity to that event rather than calibrated return levels; sensitivity to station selection is not assessed.

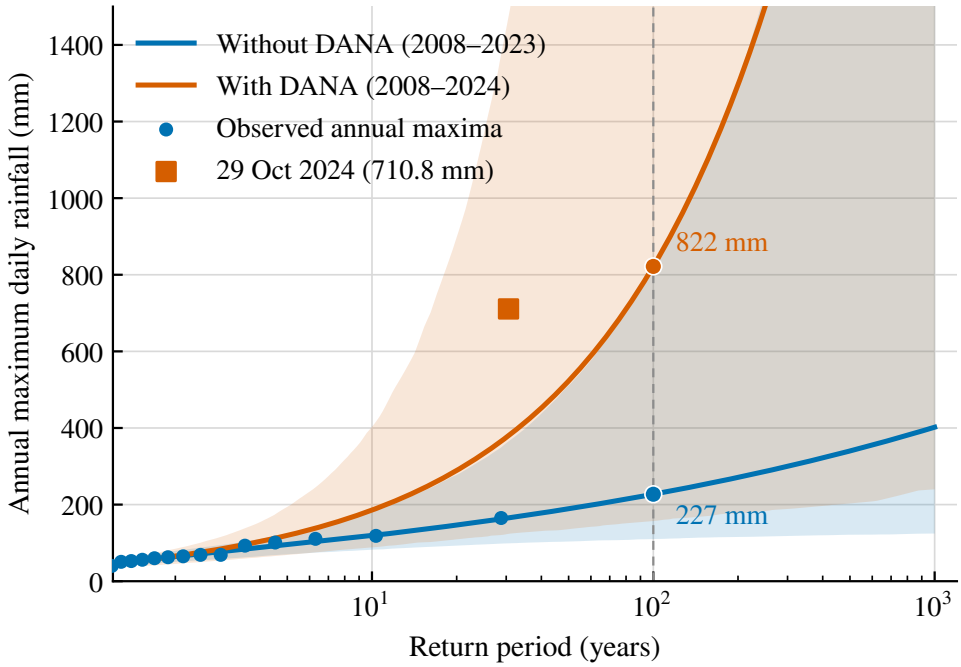


Figure 4: Turís (8337X): GEV return-level curves fitted by maximum likelihood without (16 years) and with (17 years) the 29 October 2024 event; bands: 95% Fisher intervals; dots: empirical positions (square: the 2024 record, 710.8 mm). The shape parameter rises from ξ=0.22 to 0.67 and the T=100-year level from 227 to 822 mm (table 4). The plotted values are those printed by the notebooks.

Table 4: T=100-year return level (mm) at Turís (8337X), without and with the 29 October 2024 event.

Scale	Method	Without	With	Ratio
Point	Bayesian/MCMC	258.5	932.6	3.6
Point	L-moments	218.0	755.6	3.5
Point	MLE	227.0	821.8	3.6
Regional (RFA)	L-moments	252.1	1,012.3	4.0
Regional (RFA)	MLE	271.3	1,241.3	4.6

SFINCS–HEC-RAS uncertainty comparison (Besaya River). `generate_manning_combinations` samples 1,000 roughness realisations across nine land-use classes from literature distributions. The same ensemble drives SFINCS and HEC-RAS; `load_sfincs_ensemble`, `load_flood_ensemble`, `spatial_stats` and `manning_flood_regression` aggregate inundation outputs. Figures 5 and 6 demonstrate generation and aggregation for one engine; the scientific comparison between engines is reported separately. Terrain, meshes, 1,000 outputs per engine and derived tables are openly archived [24].

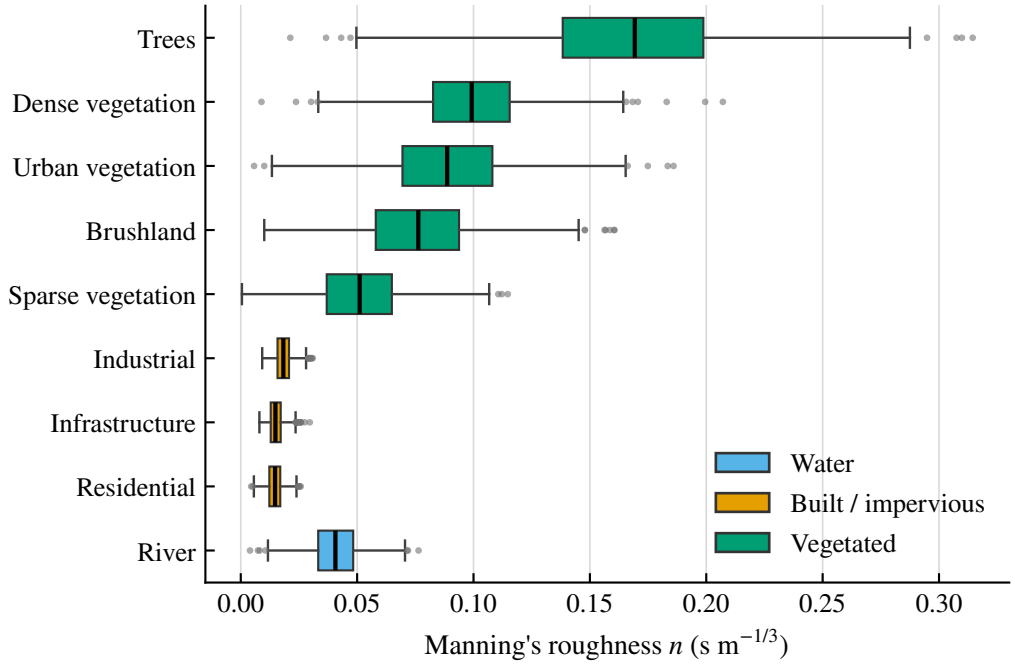


Figure 5: The 1,000 Manning combinations across nine classes, grouped by land cover and ordered by median. Boxes: interquartile range (IQR); lines: medians; whiskers: extreme values within 1.5 IQR; dots: values beyond. 995 combinations produced valid runs; the river class controls the flooded area (fig. 6). Source: [24].

These cases cover continuous impact-frequency analysis, multivariate flood frequency, extreme-rainfall inference and roughness uncertainty, using shared components.

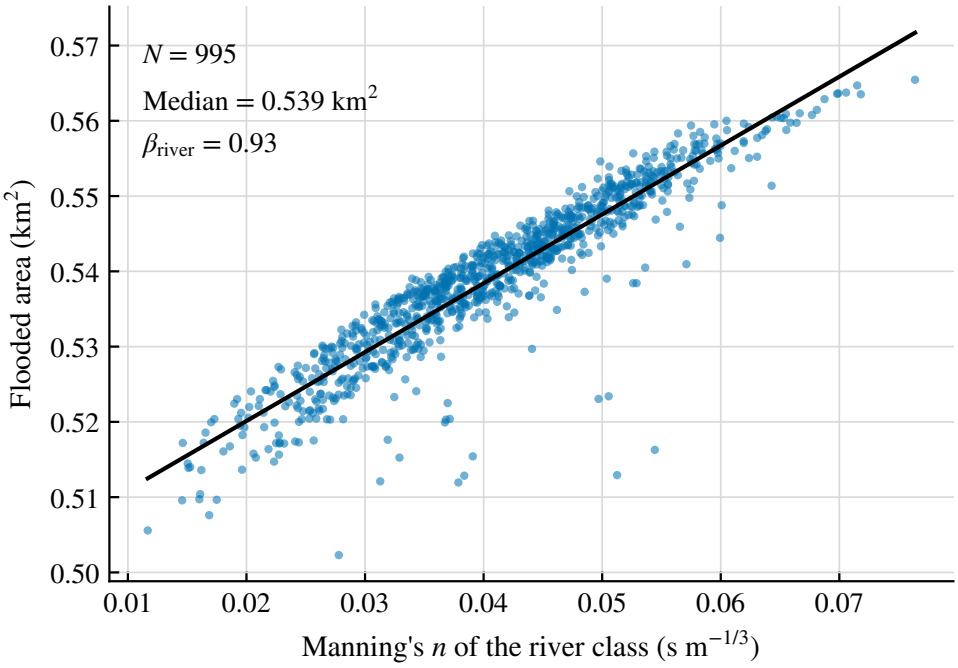


Figure 6: Output of the same case: flooded area of each of the 995 valid SFINCS runs (five numerically non-converged runs removed) against the Manning n of the river class, from the archived per-run results [24]; line: least-squares fit. In a regression on the nine standardised classes the river carries β=0.93 and every other class $|\beta| \leq 0.08$ (R^2=0.87). This single-engine result documents the ensemble-aggregation workflow only; the two-engine comparison is out of scope here and reported separately.

4. Impact

Six journal articles from the authors' group use methods consolidated in **pyhydra** [2–7]. This track record predates the software product and supports the underlying methods; it does not establish external software adoption.

The four workflows now share reusable components and ensemble/impact statistics instead of independent scripts. This supports investigating how engines propagate identical roughness ensembles and how impact-based frequency estimates differ from univariate estimates. The former comparison is reported separately; the latter is quantified by the published M-30 study. Applying the generic copula fitter to M-30 also exposed inappropriate heavy-tailed marginal selection for zero-inflated rainfall. Restricted families and zero-inflated marginals are now included in the tested API.

Zenodo archives provide citable versions of **pyhydra** [27] and HYDRA [28]. HYDRA's README gives a browser deployment, reachable on 24 October

2026, with the scientific stack pre-installed.[1] Use remains within the authors' research group at IHCantabria–Universidad de Cantabria. Independent downloads, contributions, citations, teaching and commercial adoption are not yet evidenced.

5. Conclusions

pyhydra integrates hydro-climatic data access, statistical analysis, model automation and UQ in a modular GPL-3.0-only Python library. HYDRA adds optional browser and container access. Four workflows demonstrate reuse across fluvial, urban and extreme-rainfall studies. Archived M-30 design depths and HEC-HMS/HEC-RAS results are not validation of the library, and external adoption remains to be demonstrated.
Short records constrain extreme-value inference; RFA requires homogeneous regions; stochastic generators do not resolve climate non-stationarity; and emulation depends on representative designs and physical-model validation. External-engine versions, licences and project/data availability condition reproducibility. Future work will expand test coverage and model adapters and document a stable public API for external contributors.

CRediT authorship contribution statement

Salvador Navas: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Manuel del Jesus:** Conceptualization, Methodology, Resources, Supervision, Funding acquisition, Validation, Writing – review & editing.

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[1] The Azure Container Apps deployment is ephemeral; Zenodo provides the durable scholarly records.

Declaration of competing interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Manuel del Jesus reports financial support was provided by Spain Ministry of Science Innovation and Universities. Salvador Navas reports financial support was provided by Santander Bank SA. Salvador Navas reports a relationship with Hidralab Ingeniería y Desarrollos, S.L. that includes: employment and equity or stocks. Salvador Navas reports a previous relationship with IH Cantabria that included: employment, until June 2025.

Data availability

Versioned code records are given in table 1. The independent MIT-licensed HYDRA platform is archived separately [28]. Besaya projects and processed data, the M-30 synthetic sample and archived results, and Valencia series/results are deposited under CC-BY-4.0 [23]. SFINCS–HEC-RAS configurations, outputs, tables and scripts have a separate archive [24]. The 1,761 historical M-30 events are not redistributed; engines must be obtained from their developers. Besaya includes series derived from AEMET and the Confederación Hidrográfica del Cantábrico; Valencia uses AEMET, SIAR and AVAMET. These third-party records are not offered as open data: their reuse and redistribution remain subject to provider terms.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation the authors used ChatGPT and Codex (OpenAI) and Claude Sonnet 5, Claude Opus 5.5 and Claude Sonnet 5.5 (Anthropic) for language editing, text revision, formatting, figure and plotting code, running and fixing the software and notebooks, and compiling the manuscript. The tools did not design the methodology or validate its scientific content. The authors reviewed and edited the output, checked the cited evidence and numerical values, and take full responsibility for the content of the article and the archived code.

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(orthophotography, digital terrain models, land cover and other mapping). Cloud execution and deployment of HYDRA on Microsoft Azure were supported by the Banco Santander scholarship acknowledged above.

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Current executable software version

pyhydra v0.3.1 (commit `e96fbca`) is the executable software version described in this publication; its archived record and immutable GitHub tag link are provided in Table 1 and should be cited independently using the corresponding Zenodo record [27]. The companion deployment platform HYDRA used for the browser-accessible examples in section 3 is archived separately at v0.2.2 (commit `3f5515f`) [28].